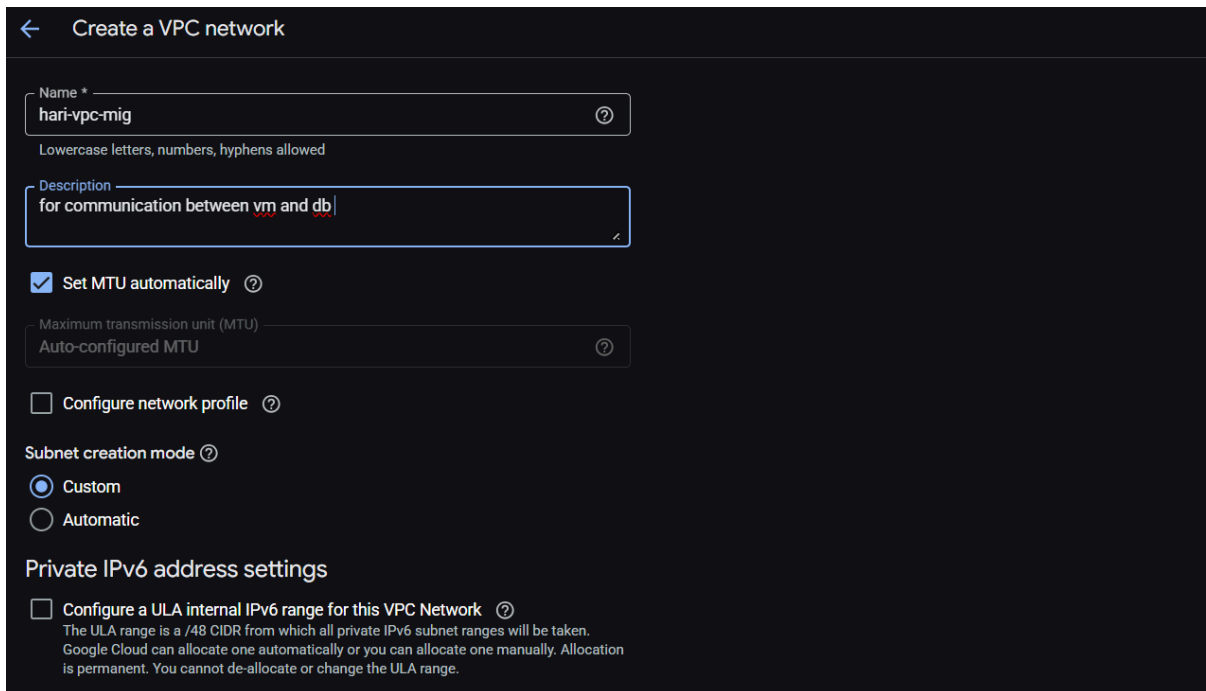


Reverse ssh method

Step 1:

- First need to create on vpc in GCP



← Create a VPC network

Name *
hari-vpc-mig ⓘ
Lowercase letters, numbers, hyphens allowed

Description
for communication between vm and db ⓘ

☒ Set MTU automatically ⓘ

Maximum transmission unit (MTU)
Auto-configured MTU ⓘ

☐ Configure network profile ⓘ

Subnet creation mode ⓘ

☒ Custom
☐ Automatic

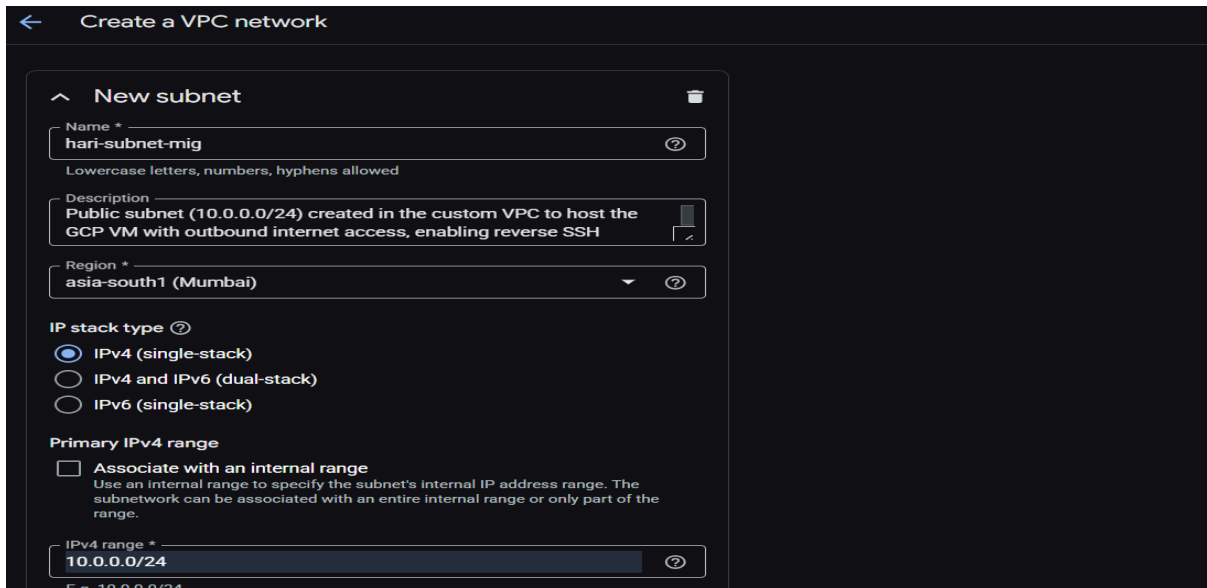
Private IPv6 address settings

☐ Configure a ULA internal IPv6 range for this VPC Network ⓘ
The ULA range is a /48 CIDR from which all private IPv6 subnet ranges will be taken. Google Cloud can allocate one automatically or you can allocate one manually. Allocation is permanent. You cannot de-allocate or change the ULA range.

- Vpc name: hari-vpc-mig
- Subnet creation mode : custom

Step 2:

- Next need to create subnet



Create a VPC network

New subnet

Name *
hari-subnet-mig

Description
Public subnet (10.0.0.0/24) created in the custom VPC to host the GCP VM with outbound internet access, enabling reverse SSH

Region *
asia-south1 (Mumbai)

IP stack type

- ☒ IPv4 (single-stack)
- ☐ IPv4 and IPv6 (dual-stack)
- ☐ IPv6 (single-stack)

Primary IPv4 range

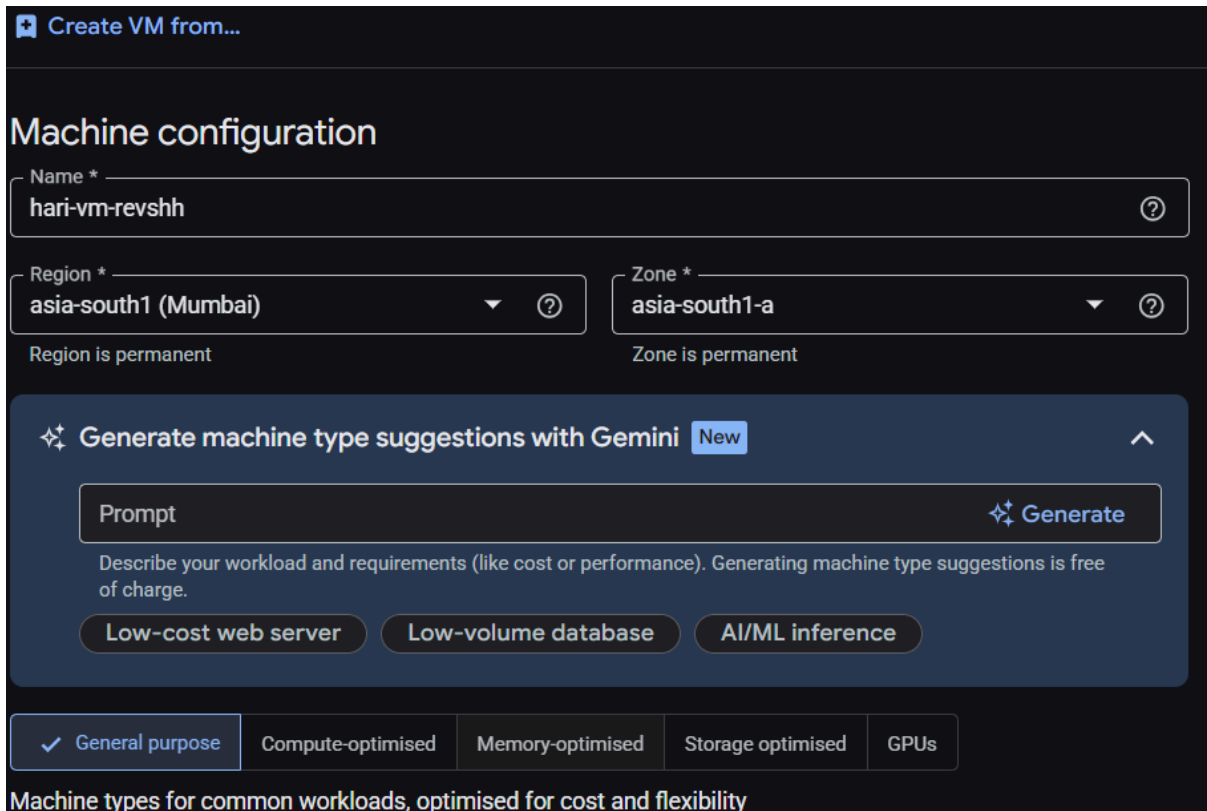
☐ Associate with an internal range

IPv4 range *
10.0.0.0/24

- subnet name: hari-subnet-mig
- Ipv4 range: 10.0.0.0/24

Step 3:

- Create on vm instance to make reverse ssh on that



Create VM from...

Machine configuration

Name *
hari-vm-revssh

Region *
asia-south1 (Mumbai)

Zone *
asia-south1-a

Region is permanent

Zone is permanent

Generate machine type suggestions with Gemini New

Prompt

Describe your workload and requirements (like cost or performance). Generating machine type suggestions is free of charge.

Low-cost web server Low-volume database AI/ML inference

☒ General purpose Compute-optimised Memory-optimised Storage optimised GPUs

Machine types for common workloads, optimised for cost and flexibility

- Vm name: hari-vm-revssh

- Region: asia-south1 (mumbai)
- Zone: asia-south1-a
- Machine type: e2-micro
- Image: Debian GNU/Linux 12 (bookworm)

Operating system and storage

Name	hari-vm-revshh
Type	New balanced persistent disk
Size	10 GB
Snapshot schedule [?]	No schedule selected
Licence type [?]	Free
Image	 Debian GNU/Linux 12 (bookworm)

- Default only os and storage

^ Edit network interface

Network *
hari-vpc-mig  

Subnetwork *
hari-subnet-mig IPv4 (10.0.0.0/24)  

 To use IPv6, you need an IPv6 subnet range.
[Learn more](#) 

Network interface card
—  

IP stack type
☒ IPv4 (single-stack)
☐ IPv4 and IPv6 (dual-stack)
☐ IPv6 (single-stack)

Primary internal IPv4 address
Ephemeral (Automatic)  

- In vm we need to choose the network and subnetwork which we created and then click create finally.

Step 4:

- Create one db on cloud sql postgresql engine as private

Instance info

Database version *
PostgreSQL 18

Instance ID *
postgres
Use lowercase letters, numbers and hyphens. Start with a letter.

Password *

[Generate](#)

Set a password for the default admin user postgres. [Learn more](#)

- ✓ Must contain an uppercase letter, lowercase letter, number and non-alphanumeric character
- ✓ Must not contain the username
- ✓ Must be at least 8 characters long

✓ Password policy

Choose region and zonal availability

For better performance, keep your data close to the services that need it. Region is permanent, while zone can be changed any time.

Region
asia-south1 (Mumbai)

Zonal availability
☒ Single zone
In case of outage, no failover. Not recommended for production.

Choose a private connection method ⓘ

☒ Private services access (PSA)
For connecting to single VPC networks.

VPC network *
hari-vpc-mig

❗ Network connection is not set up

Allocated IP range
Automatic ⓘ

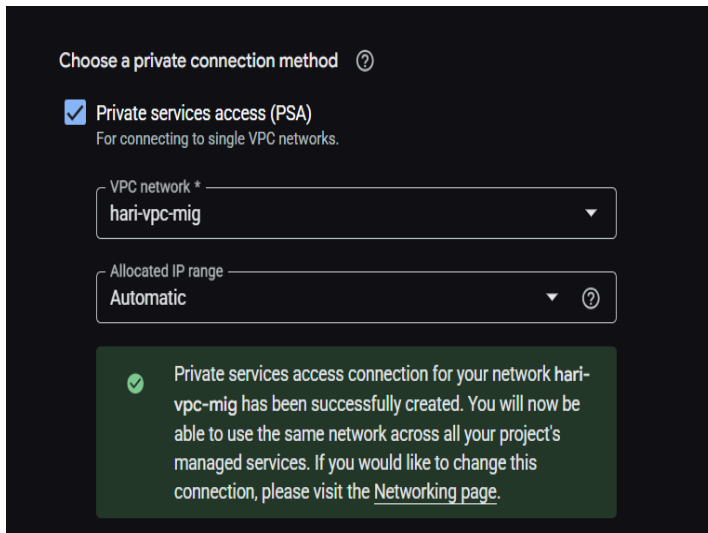
Network setup confirmation required

Your network hari-vpc-mig requires a private services access connection. This connection enables your services to communicate exclusively by using internal IP addresses. [Learn more](#)

This is a one-time setup per VPC. It typically takes 1 to 2 minutes.

✓ Setting up connection (elapsed time: 1 sec)

- Select engine: postgres 18
- Db id: postgres
- Password: ABC!123abc? (need to create or generate)
- Region: asia-south1 (mumbai)
- Zone: single zone
- Machine & storage config: choose minimum level things
- Network connections: select the created vpc (hari-vpc-mig) and create one private service account for this by click setup



- Then finally click create instance.

Step 5:

Need to create on vpc in aws

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only
 ☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block [Info](#)

☒ IPv4 CIDR manual input
 ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

CIDR block size must be between /16 and /28.

- Vpc name: hari-vpc-mig
- Ivp4 range: 10.0.0.0/16

Step 6:

- Need to create two subnets under this vpc as public and private

Subnet 1 of 2**Subnet name**

Create a tag with a key of 'Name' and a value that you specify.

hari-private-subnet

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1a

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.1.0/24

256 IPs

- Subnet name: hari
- Zone: asia pacific (mumbai) - (ap-south-1a)
- Ipv4 block: 10.0.1.0/24

- Important one, whenever creating the subnets assign the zone as different like public subnet as ap-south1-a and private subnet as ap-south1-b

Subnet 2 of 2**Subnet name**

Create a tag with a key of 'Name' and a value that you specify.

hari-public-subnet

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1b

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.2.0/24

256 IPs

- Subnet name: hari-public
- Zone: asia pacific (mumbai) - (ap-south-1b)
- Ipv4 block: 10.0.2.0/24

Step 7:

- Create one internet gateway and attach that to the vpc which you created.

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key



Value - optional



You can add 49 more tags.

igw-0a196ad346f2e03ff / hari-ig-mig

Details [Info](#)

Internet gateway ID

State

VPC ID

Owner

Step 8:

- Create on NAT for vm instance to access internet
- NAT name: hari-nat-vm
- Vpc: which we created (hari-vpc-mig)

NAT gateway settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability mode [Info](#)

Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☒ **Regional - new**

Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☐ **Zonal**

Provides granular control within a specific availability zone, adhering to subnet level settings.

VPC

Select a VPC in which to create the regional NAT gateway.

Connectivity type

Select a connectivity type for the NAT gateway.

☒ **Public**
☐ Private

Step 9:

- Create private route table

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings**Name - optional**

Create a tag with a key of 'Name' and a value that you specify.

VPC

The VPC to use for this route table.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

You can add 49 more tags.

- Route table name: hari-route-private
- Vpc: hari-vpc-mig
- After creating the route table go to the subnet associations and click on edit subnet associations and select the subnet which is mentioned private

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	hari-private-subnet	subnet-01770168a6e541a1f	10.0.1.0/24	–	Main (rtb-0aad93603bb767e54)
☐	hari-public-subnet	subnet-01c5ce430425690b2	10.0.2.0/24	–	Main (rtb-0aad93603bb767e54)

Selected subnets

- Create public route table
- Route table name: hari-route-public
- Vpc: hari-vpc-mig

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings**Name - optional**

Create a tag with a key of 'Name' and a value that you specify.

VPC

The VPC to use for this route table.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

You can add 49 more tags.

- After creating the route table go to the subnet associations and click on edit subnet associations and select the subnet which is mentioned public

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐ hari-private-subnet	subnet-01770168a6e541a1f	10.0.1.0/24	-	rtb-026df8f2868fed09b / hari-route-private
☒ hari-public-subnet	subnet-01c5ce430425690b2	10.0.2.0/24	-	Main (rtb-0aad93603bb767e54)

Selected subnets

subnet-01c5ce430425690b2 / hari-public-subnet X

Cancel Save associations

Step 10:

- In private route table click edit routes and add the nat gateway which we created

VPC > Route tables > rtb-026df8f2868fed09b > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute

Remove

Add route

Cancel Preview Save changes

- In public route table click edit routes and add the internet gateway which we created

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Remove

Add route

Cancel Preview Save changes

Step 11:

- Create one ec2

[Connect](#)
[Instance state ▾](#)
[Actions ▾](#)

Instance ID
 i-Oea54f99343e1a2a6

IPv6 address
 -

Hostname type
 IP name: ip-10-0-2-68.ap-south-1.compute.internal

Answer private resource DNS name
 -

Auto-assigned IP address
 3.7.66.84 [Public IP]

IAM role
 -

Public IPv4 address
 3.7.66.84 | [open address ↗](#)

Instance state
 **Running**

Private IP DNS name (IPv4 only)
 ip-10-0-2-68.ap-south-1.compute.internal

Instance type
 t2.nano

VPC ID
 vpc-019573dfcb9b7ace3 (hari-vpc-mig) [↗](#)

Subnet ID
 subnet-01c5ce430425690b2 (hari-public-subnet) [↗](#)

Private IPv4 addresses
 10.0.2.68

Public DNS
 —

Elastic IP addresses
 —

AWS Compute Optimizer finding
 Opt-in to AWS Compute Optimizer for recommendation.
[Learn more](#) 

Auto Scaling Group name
 —

- Name of ec2: hari-ec2-revssh
- Image: ubuntu
- Machine: minimum leve nano

▼ Network settings

Info

VPC - required

Info

vpc-019573dfcb9b7ace3 (hari-vpc-mig)

10.0.0.0/16

↻

Subnet

Info

subnet-01c5ce430425690b2

hari-public-subnet

↻ Create new subnet

VPC: vpc-019573dfcb9b7ace3

Owner: 317968578233

Availability Zone: ap-south-1b (aps1-az3)

Zone type: Availability Zone

IP addresses available: 251

CIDR: 10.0.2.0/24

Auto-assign public IP

Info

Enable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

hari-ec2-mig-security-group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./!@#.%&()*~`

Description - required

Info

created 2026-04-15T09:08:00.255Z

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-05d2d839d4f73aafb

Virtual server type (instance type)

t2.nano

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

- Vpc: hari-vpc-mig
 - Subnet: hari-subnet-public
 - Security group: create one new
- In the security we need to edit and now fix the ssh as 22 port with tcp default one.
- After that create one new key pair to access the vm by this key via ssh any where, for that select the RSA type and select pem file type and enter the name of the key
- Name of key: hari-ec2-mig
 - File type: pem
 - Key type: RSA

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Cancel
Create key pair

Step 12:

- Create one subnet group on database

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name
You won't be able to modify the name after your subnet group has been created.

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

▼

2 Subnets, 2 Availability Zones

- Subnet group name: hari-db-subnet
- Vpc: hari-vpc-mig

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

ap-south-1a ✕ ap-south-1b ✕ ap-south-1c ✕

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

hari-private-subnet ✕ Subnet ID: subnet-01770168a6e541a1f CIDR: 10.0.1.0/24

hari-public-subnet ✕ Subnet ID: subnet-01c5ce430425690b2 CIDR: 10.0.2.0/24

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (2)

Availability zone	Subnet name	Subnet ID	CIDR block
ap-south-1a	hari-private-subnet	subnet-01770168a6e541a1f	10.0.1.0/24
ap-south-1b	hari-public-subnet	subnet-01c5ce430425690b2	10.0.2.0/24

Subnet: private subnet which created and select the zone availability all three

Step 13:

- Create the RDS with full configuration

Aurora and RDS > Databases > Create database

Create database Info

Engine options

Engine type Info

☐ Aurora (MySQL Compatible) 

☐ Aurora (PostgreSQL Compatible) 

☐ MySQL 

☒ PostgreSQL 

☐ MariaDB 

☐ Oracle 

☐ Microsoft SQL Server 

☐ IBM Db2 

Choose a database creation method

☒ Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

- Select the postgresql engine and full configuration
- Select the sandbox option and subnet group which we created

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

hari-vpc-mig (vpc-019573dfcb9b7ace3)
2 Subnets, 2 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

hari-db-subnet
2 Subnets, 2 Availability Zones

Public access [Info](#)

☐ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☐ Choose existing
Choose existing VPC security groups

☒ **Create new**
Create new VPC security group

New VPC security group name
hari-db-security-group

- Select the vpc and subnet both and disable the public access
- Create one new security group with name

Edit inbound rules [Info](#)
Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sgr-07671575ba6a89e1e	PostgreSQL	TCP	5432	Custom	3.7.66.84/32	Delete

[Add rule](#)

[Cancel](#) [Preview changes](#) [Save rules](#)

- In the security group mention the ec2 public ip address

Step 14:

From kaggle we downloaded the sample databases, we uploaded the databases on bucket s3. Now we need to dump that db into postgres db which we created, for that to connect to ec2 via ssh and put the endpoint of db to get access to db then copy the db to ec2 and dump into the postgres which we have.

EC2 Instance Connect | SSM Session Manager | SSH client | EC2 serial console

Instance ID
i-Oea54f99343e1a2a6 (hari-ec2-revssh)

Connection type

☒ **Connect using a Public IP**
Connect using a public IPv4 or IPv6 address

☐ **Connect using a Private IP**
Connect using a private IP address and a VPC endpoint

☒ **Public IPv4 address**
3.7.66.84

☐ **IPv6 address**

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

ubuntu

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

[Cancel](#) [Connect](#)

Check PostgreSQL client

Shell

```
sudo apt update
sudo apt install wget gnupg -y

echo "deb http://apt.postgresql.org/pub/repos/apt $(lsb_release -cs)-pgdg main"
| sudo tee /etc/apt/sources.list.d/pgdg.list

wget -qO - https://www.postgresql.org/media/keys/ACCC4CF8.asc | sudo apt-key
add -

sudo apt update
sudo apt install postgresql-client-17 -y
```

Check dump file exists

Shell

```
ls
```

Should show:
testdb.dump

Identify dump type
file testdb.dump

Output:

PostgreSQL custom database dump

Shell

```
Connect to RDS
psql -h postgres.czsgeaiqe50i.ap-south-1.rds.amazonaws.com -U postgres -d
postgres
```

SQL

Create target DB

```
CREATE DATABASE testdb;
```

```
SQL  
Exit:  
\q
```

Restore dump (IMPORTANT STEP)

```
Shell  
pg_restore -h postgres.czsgeaiqe50i.ap-south-1.rds.amazonaws.com \  
-U postgres \  
-d testdb \  
--no-owner \  
--no-acl \  
Testdb.dump  
  
Enter password
```

```
Shell  
Connect to restored DB  
  
psql -h postgres.czsgeaiqe50i.ap-south-1.rds.amazonaws.com -U postgres -d  
testdb
```

```
SQL  
Verify tables  
  
\dt
```

```
SQL  
Output:  
public | stores | table | postgres
```

SQL

Verify data

```
SELECT * FROM stores LIMIT 5;
```

SQL

Output:

S001		Chocolate Store 1		New York		Canada		Retail
S002		Chocolate Store 2		Melbourne		Canada		Mall
S003		Chocolate Store 3		Berlin		France		Mall
S004		Chocolate Store 4		Paris		UK		Airport
S005		Chocolate Store 5		Sydney		USA		Online

```
ubuntu@ip-10-0-2-68:~$ sudo apt update
sudo apt install wget gnupg -y
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Hit:9 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7364 B]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 852 kB in 1s (1099 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
47 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
gnupg is already the newest version (2.4.4-2ubuntu17.4).
gnupg set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 47 not upgraded.
ubuntu@ip-10-0-2-68:~$ echo "deb http://apt.postgresql.org/pub/repos/apt $(lsb release -cs)-pgdg main" | sudo tee /etc/apt/sources.list.d/pgdg.list
```

```
ubuntu@ip-10-0-2-68:~$ pg_restore -h postgres.czsgaeiqe50i.ap-south-1.rds.amazonaws.com \
-U postgres \
-d testdb \
--no-owner \
--no-acl \
testdb.dump
Password:
ubuntu@ip-10-0-2-68:~$ psql -h postgres.czsgaeiqe50i.ap-south-1.rds.amazonaws.com -U postgres -d testdb
Password for user postgres:
```



```

testdb-> \q
ubuntu@ip-10-0-2-68:~$ psql -h postgres.czsgeaiqe50i.ap-south-1.rds.amazonaws.com -U postgres -d testdb
Password for user postgres:
psql (17.9 (Ubuntu 17.9-1.pgdg24.04+1), server 18.1)
WARNING: psql major version 17, server major version 18.
        Some psql features might not work.
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off, ALPN: postgresql)
Type "help" for help.

testdb-> \dt
      List of relations
 Schema | Name  | Type  | Owner
-----+-----+-----+-----
 public | stores | table | postgres
(1 row)

testdb-> SELECT * FROM stores LIMIT 5;
 store_id | store_name | category | city | state
-----+-----+-----+-----+-----
 S001     | Chocolate Store 1 | New York | Canada | Retail
 S002     | Chocolate Store 2 | Melbourne | Canada | Mall
 S003     | Chocolate Store 3 | Berlin | France | Mall
 S004     | Chocolate Store 4 | Paris | UK | Airport
 S005     | Chocolate Store 5 | Sydney | USA | Online
(5 rows)

testdb->

```

Step 15:

Editing the security groups which we created RDS and EC2

RDS Security Group:

Allow:

- Type: PostgreSQL
- Port: 5432
- Source: EC2 PRIVATE IP (10.0.x.x)

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
sgr-07671575ba6a89e1e	PostgreSQL	TCP	5432	Custom	Q 10.0.2.68/32 10.0.2.68/32 Delete

EC2 Security Group:

Allow:

- SSH → 22 → 0.0.0.0/0 or VM ip
- OUTBOUND → ALL TRAFFIC (important for reverse SSH)

Edit inbound rules [Info](#)
Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sgr-0499290c31966d40d	SSH	TCP	22	My IP		Delete
Add rule						

223.185.21.127/32 [X](#)

[Cancel](#) [Preview changes](#) [Save rules](#)

Step 16:

- Create the firewall rule on gcp

Create a firewall rule

☒ Allow
☐ Deny

Targets
All instances in the network

Source filter
IPv4 ranges

Source IPv4 ranges *
0.0.0.0/0 for example, 0.0.0.0/0, 192.168.2.0/24

Second source filter
None

Destination filter
None

Protocols and ports
☐ Allow all
☒ Specified protocols and ports

☒ TCP
Ports
5433,22
E.g. 20, 50-60

- Name : allow-tunnel-reversessh
- Source: 0.0.0.0/0
- TCP port: 5433,22
- After that open the ssh of the vm and check with this command

```
Shell
sudo nano /etc/ssh/sshd_config
```

Change the gatewayport as yes and allowtcpforwarding as yes

```

GNU nano 7.2 /etc/ssh/sshd_config *
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes

AllowAgentForwarding yes
AllowTcpForwarding yes
GatewayPorts yes
X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem        sftp    /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line  M-E Redo

```

Then restart the ssh by this command

```

Shell
sudo systemctl restart ssh

```

Then on aws connect ssh gcp by these command

```

ubuntu@ip-10-0-2-68:~$ ssh -i gcp-key -v hariharan_j_cloudambassadors_com@34.14.134.71
OpenSSH_9.6p1 Ubuntu-3ubuntu13.15, OpenSSL 3.0.13 30 Jan 2024
debug1: Reading configuration data /etc/ssh/ssh_config
debug1: /etc/ssh/ssh_config line 19: include /etc/ssh/ssh_config.d/*.conf matched no files
debug1: /etc/ssh/ssh_config line 21: Applying options for *
debug1: Connecting to 34.14.134.71 [34.14.134.71] port 22.
debug1: Connection established.
debug1: identity file gcp-key type 0
debug1: identity file gcp-key-cert type -1
debug1: Local version string SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.15
debug1: Remote protocol version 2.0, remote software version OpenSSH_9.2p1 Debian-2+deb12u9
debug1: compat_banner: match: OpenSSH_9.2p1 Debian-2+deb12u9 pat OpenSSH* compat 0x04000000
debug1: Authenticating to 34.14.134.71:22 as 'hariharan_j_cloudambassadors_com'
debug1: load_hostkeys: fopen /home/ubuntu/.ssh/known_hosts2: No such file or directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts: No such file or directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts2: No such file or directory
debug1: SSH2_MSG_KEXINIT sent
debug1: SSH2_MSG_KEXINIT received
debug1: kex: algorithm: sntrup761x25519-sha512@openssh.com
debug1: kex: host key algorithm: ssh-ed25519

```

```

nlistp256@openssh.com,weauthn-sk-ecdsa-sha2-nistp256@openssh.com,ssh-dss,ssh-rsa,rsa-sha2-256,rsa-sha2-512>
Debug1: key_ext_info_check_ver: publickey-hostbound@openssh.com=<0>
Debug1: SSH2 MSG SERVICE ACCEPT received
Debug1: Authentications that can continue: publickey
Debug1: Next authentication method: publickey
Debug1: Will attempt key: gcp-key RSA SHA256:54eysaVzQI+qDkrlTxRL+BQeDjSMqdBmLP+1XiH6zpo explicit
Debug1: Offering public key: gcp-key RSA SHA256:54eysaVzQI+qDkrlTxRL+BQeDjSMqdBmLP+1XiH6zpo explicit
Debug1: Authentications that can continue: publickey
Debug1: No more authentication methods to try.
hariharan_j_cloudambassadors_com@34.14.134.71: Permission denied (publickey).
ubuntu@ip-10-0-2-68:~$ ssh -i gcp-key hariharan_j_cloudambassadors_com@34.14.134.71
Linux hari-vm-revssh 6.1.0-44-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.164-1 (2026-03-09) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Apr 15 12:02:04 2026 from 35.235.243.130
hariharan_j_cloudambassadors_com@hari-vm-revssh:~$ ssh -i gcp-key -N -R 5433:postgres.czsgaeiqe50i.ap-south-1.rds.amazonaws.com:5432 hariharan_j_cloudambassadors_c
om@34.14.134.71
The authenticity of host '34.14.134.71 (34.14.134.71)' can't be established.
ED25519 key fingerprint is SHA256:1LfrY6wifwtAkjgDsmAh/VLVGmbFVS20mH6cMGYa3Kw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '34.14.134.71' (ED25519) to the list of known hosts.

```

GCP side verification

```

root@hari-vm-revssh:~# ss -tulnp | grep 5433
tcp    LISTEN 0      128          0.0.0.0:5433          0.0.0.0:*        users:(("sshd",pid=21781,fd=7))
tcp    LISTEN 0      128          [::]:5433          [::]:*           users:(("sshd",pid=21781,fd=9))
root@hari-vm-revssh:~#

root@hari-vm-revssh:~# psql -h localhost -p 5433 -U postgres -d postgres
Password for user postgres:
psql (15.16 (Debian 15.16-0+deb12u1), server 18.1)
WARNING: psql major version 15, server major version 18.
         Some psql features might not work.
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off)
Type "help" for help.

postgres=>

```

None

EC2 (AWS) → RDS → (Reverse SSH Tunnel) → GCP VM → DMS

GCP VM SETUP (DO FIRST)

Login to GCP VM:

Shell

`ssh hariharan_j_cloudambassadors_com@34.14.134.71`

Shell

`sudo nano /etc/ssh/sshd_config`

Make sure these lines exist at bottom:

```
Shell
PubkeyAuthentication yes
AuthorizedKeysFile .ssh/authorized_keys
AllowTcpForwarding yes
GatewayPorts yes
PermitTunnel yes
```

ALSO comment these lines (VERY IMPORTANT):

```
Shell
#AuthorizedKeysCommand /usr/bin/google_authorized_keys
#AuthorizedKeysCommandUser root
```

Restart SSH

```
Shell
sudo systemctl restart ssh
```

Create SSH key (for tunnel)

```
Shell
ssh-keygen -t rsa -f gcp-key
```

Press ENTER 3 times

Add public key

```
Shell
mkdir -p ~/.ssh
nano ~/.ssh/authorized_keys
```

Paste:

```
None
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQ...
```

(save + exit) by ctrl o and enter and ctrl x

Fix permissions

```
Shell
chmod 700 ~/.ssh
```

```
chmod 600 ~/.ssh/authorized_keys
```

6. Open firewall (GCP)

Allow port **5433**

```
Shell
gcloud compute firewall-rules create allow-5433 \
  --allow tcp:5433 \
  --source-ranges 0.0.0.0/0 \
  --target-tags allow-5433
```

AWS EC2 SETUP

Login to EC2:

```
Shell
ssh -i hari-ec2-mig.pem ubuntu@3.7.66.84
```

Check RDS connection

```
Shell
psql -h postgres.czsgeaiqe50i.ap-south-1.rds.amazonaws.com -U postgres -d postgres
```

Must connect

Copy GCP private key (VERY IMPORTANT)

From GCP → EC2:

```
Shell
scp -i hari-ec2-mig.pem gcp-key ubuntu@3.7.66.84:/home/ubuntu/
```

Fix permissions on EC2

```
Shell  
chmod 400 gcp-key
```

Run reverse SSH tunnel

```
Shell  
ssh -i gcp-key -N -R 0.0.0.0:5433:postgres.czsgeaiqe50i.ap-south-1.rds.amazonaws.com:5432  
hariharan_j_cloudambassadors_com@34.14.134.71
```

KEEP THIS TERMINAL OPEN

VERIFY FROM GCP

Open new terminal → login GCP:

```
Shell  
ssh hariharan_j_cloudambassadors_com@34.14.134.71
```

Test tunnel

```
Shell  
psql -h localhost -p 5433 -U postgres -d postgres
```

Expected:

```
None  
postgres=>
```

TROUBLESHOOT (IF FAILS)

Check tunnel running (on GCP)

```
Shell  
ss -tulnp | grep 5433
```

Must show:

None

0.0.0.0:5433 LISTEN

If connection fails:

Check EC2 still running tunnel

Check RDS security group

Check GCP firewall

FINAL RESULT

None

GCP VM:5433 → EC2 → RDS

Step 17:

- Creating the dms on gcp

← Create a migration job

1 Get started
Not configured

2 Define a source
Not configured

3 Define a destination
Not configured

4 Define connectivity method
Not configured

5 Configure migration databases
Not configured

6 Test and create migration job
Not tested

Save and Exit Cancel

Describe your migration job

Provide some basic info about your migration job and review what you need to do to complete it successfully. Want to know which types of migrations are supported right now? [Learn more](#)

Note: All DMS resources must share the same region as the destination database. This includes the conversion workspace, connection profiles, migration job and private connectivity configuration (if relevant).

Migration job name *
hari-aws-gcp-reversessh
Must be fewer than 60 characters. 23/60

Migration job ID *
hari-aws-gcp-reversessh
Lowercase letters, numbers or hyphens. It must be unique in this project, and cannot be changed later. 23/60

Source database engine *
Amazon RDS for PostgreSQL

Destination database engine *
Cloud SQL for PostgreSQL

Destination region *
asia-south1 (Mumbai)
Choice is permanent. For performance, choose a region that is close to the services that need your data.

Migration job type *
Continuous

- Migration job name: hari-aws-gcp-reversessh
- Source: amazon rds for postgresql
- Destination: cloud sql for postgresql
- Region: asia-south1 (mumbai)
- Migration job type: continuous

Google Cloud

internal-sandbox

Search (/) for resources, docs, products and more

Database Migration / Migration jobs / Create migration

Create a migration job

Get started

hari-aws-gcp-reversesh, Amazon RDS for PostgreSQL to Cloud SQL for PostgreSQL (Continuous)

Define a source

Not configured

Define a destination

Not configured

Define connectivity method

Not configured

Configure migration databases

Not configured

Test and create migration job

Not tested

Save and Exit Cancel

Define your source

Your data source's connection profile represents all the info needed to connect to your source database.

Choose a connection profile that already exists. [Learn more](#)

Source database engine

Amazon RDS for PostgreSQL

Select source connection profile *

Filter Type to filter

aws-gcp-private-db

aws-source

debug-test

dwadd

harsha-profile

hsb-profile

hsb-rds-source

Create a connection profile

PostgreSQL to PostgreSQL

Configure this PostgreSQL source connection profile for a migration with Cloud SQL for PostgreSQL and AlloyDB.

Define connection details

Enter connection details for your connection profile.

Hostname or IP address *

34.14.134.71

Port *

5433

Username *

postgres

Password *

Secure your connection

Choose an encryption type, and you'll see the SSL/TLS details needed.

Encryption type *

None

Save Cancel

Create a migration job

Get started

hari-aws-gcp-reversesh, Amazon RDS for PostgreSQL to Cloud SQL for PostgreSQL (Continuous)

Define a source

hari-aws-db

Define a destination

Not configured

Define connectivity method

Not configured

Configure migration databases

Not configured

Test and create migration job

Not tested

Save and Exit Cancel

Define your destination

You can create a new instance or select an existing instance to be associated with this migration job.

Under-provisioned storage and compute resources can slow down your migration job. For a smooth database migration, ensure that your instance has adequate machine and storage configurations to accommodate the size of your source database. [Learn more](#)

Destination database engine

Cloud SQL for PostgreSQL

Type of destination instance *

Existing instance

This choice is permanent.

DMS will demote the existing instance and convert it into a replica for the source database. Only empty instances are supported, otherwise the migration will fail to start. [Learn more](#)

Select destination instance

Instance ID *

postgres

Instance ID

postgres

Region

asia-south1 (Mumbai)

Database version

POSTGRES_18

Database Migration / Migration jobs / Create migration

Create a migration job

Get started

hari-aws-gcp-reversesh, Amazon RDS for PostgreSQL to Cloud SQL for PostgreSQL (Continuous)

Define a source

hari-aws-db

Define a destination

postgres

Define connectivity method

Reverse-SSH tunnel via cloud-hosted VM

Configure migration databases

Not configured

Test and create migration job

Not tested

Save and Exit Cancel

Choose the best connectivity method for your migration job based on your security and throughput requirements.

Need help choosing? [View connectivity guide](#)

Connectivity method *

Reverse-SSH tunnel via cloud-hosted VM

Selecting Reverse-SSH as the connectivity method will update the instance to turn on private IP.

Configure your destination to accept connections

Make sure that the replication connections section of the pg_hba.conf file or the security groups definitions in AWS RDS on the source database are updated to accept connections from the Cloud SQL VPC's IP address range.

Choose a virtual machine to host the tunnel

Choose or create a new Compute Engine VM instance to host the reverse-SSH tunnel needed to connect your source and destination. A script will be generated to help you set up, which you can run using gcloud on a machine that has access to your source and can be available during the entire migration. [Learn more](#)

Compute Engine VM instance *

hari-vm-revshh

SSH tunnel port *

5433

Reverse SSH + DMS Migration (AWS RDS → GCP Cloud SQL)

1. Architecture Overview

AWS RDS (Private DB)



AWS EC2 (Public IP: 65.1.3.92, Private IP: 10.0.2.136)



GCP VM (hari-reverse-ssh, Internal IP: 10.0.0.4)



GCP DMS



Cloud SQL (hari-db)

2. Problem Statement

- AWS RDS is **private**
- GCP DMS cannot connect directly
- Need secure connectivity without making DB public

Solution

Use **SSH Tunnel via EC2 → GCP VM**

3. Infrastructure Details

AWS Side

Component	Value
RDS Endpoint	postgres.czsgeaiqe50i.ap-south-1.rds.amazonaws.com
RDS Port	5432
EC2 Public IP	65.1.3.92
EC2 Private IP	10.0.2.136

GCP Side

Component	Value
VM Name	hari-reverse-ssh
Internal IP	10.0.0.4
Tunnel Port	5433

4. Security Configuration

AWS Security Group (RDS)

Added rules:

1. PostgreSQL → 5432 → Source: 10.0.2.136 (EC2)
2. PostgreSQL → 5432 → Source: 0.0.0.0/0 (temporary testing)
3. Custom TCP → 5433 → Source: 0.0.0.0/0 (tunnel)

GCP Firewall Rules

Allow:

- TCP 22 (SSH)

- TCP 5433 (Tunnel)

Source: 0.0.0.0/0

5. Database Configuration (AWS RDS)

Step 1: Enable pglogical

In Parameter Group:

`shared_preload_libraries = pglogical`

Step 2: Reboot RDS

Required because parameter is static

Step 3: Enable Extension

`CREATE EXTENSION pglogical;`

6. Reverse SSH Tunnel Setup

Final Working Command (IMPORTANT)

Run on **GCP VM (hari-reverse-ssh)**:

```
ssh -i hari-rev-ssh-aws.pem -f -N \
```

```
-R 0.0.0.0:5433:postgres.czsgeaiqe50i.ap-south-1.rds.amazonaws.com:5432 \
```

```
ubuntu@65.1.3.92
```

What this does

GCP VM (10.0.0.4:5433)

↓

AWS EC2 (65.1.3.92)

↓

RDS (5432)

Verify tunnel

```
ss -tulnp | grep 5433
```

Expected:

```
0.0.0.0:5433 LISTEN
```

7. DMS Configuration

Source Connection

- Host: 10.0.0.4
- Port: 5433
- User: postgres
- DB: testdb
- SSL: None
- IMPORTANT:

DMS connects to VM → tunnel → RDS

8. Errors Faced & Fixes (VERY IMPORTANT)

Error 1: Permission denied (publickey)

ubuntu@43.x.x.x: Permission denied (publickey)

Fix

```
chmod 400 hari-rev-ssh-aws.pem
```

Error 2: Connection timed out

ssh: connect to host 43.204.96.64 port 22: Connection timed out

Fix

- Used correct IP:

65.1.3.92 (correct EC2 public IP)

Error 3: Port already in use

bind: Address already in use

Fix

```
pkill -f "ssh.*5433"
```

Error 4: Connection refused (psql)

connection refused on localhost:5433

Cause

Tunnel not created properly

Fix

Recreate SSH tunnel manually

Error 5: no pg_hba.conf entry

no pg_hba.conf entry for host "10.0.2.136"

Fix

- Adjusted RDS access via security group
- Allowed EC2 internal IP

Error 6: pglogical not enabled

pglogical is not in shared_preload_libraries

Fix

1. Add in parameter group:

shared_preload_libraries = pglogical

2. Reboot DB

Error 7: relation does not exist

relation "stores" does not exist

Cause

Connected to wrong DB (`postgres` instead of `testdb`)

Fix

`psql -d testdb`

Error 8: DMS cannot connect

Failure connecting to source database

Fix

- Correct host:

10.0.0.4 (NOT RDS endpoint)

- Ensure tunnel running

9. Validation (CDC Testing)

Insert

```
INSERT INTO stores VALUES ('S999', 'Test Store', 'TestCity', 'TestState', 'Online');
```

Appeared in Cloud SQL

Update

```
UPDATE stores SET city = 'Chennai' WHERE store_id = 'S999';
```

Reflected in Cloud SQL

Delete

```
DELETE FROM stores WHERE store_id = 'S999';
```

Removed from Cloud SQL